CYBER UTILITY

An Integrated Cybersecurity & Mathematical Operations Toolkit

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Abstract

Cyber Utility is a web-based integrated toolkit developed to assist cybersecurity students in performing essential cryptographic, mathematical, and logical computations within a single interactive platform. The system includes modules such as cryptographic conversions, brute-force attack simulation, modular arithmetic, matrix calculations, probability tools, and logic gate simulations. The project is developed using HTML, CSS, and JavaScript and is designed to enhance cybersecurity education through practical experimentation.

Introduction

With the rapid growth of digital systems, cybersecurity education has become increasingly important. Students require practical tools to understand encryption, hashing, brute-force mechanisms, number systems, and logical operations. Cyber Utility provides an integrated platform where these concepts can be explored within a single browser-based environment.

Problem Statement

Cybersecurity students often use multiple disconnected tools for cryptographic and mathematical operations. This creates inefficiency and limits integrated learning. There was a need for a unified educational toolkit that combines essential cybersecurity functions in one interactive platform.

Objectives

• Develop an integrated cybersecurity utility platform.

• Simulate brute-force attack mechanisms for educational purposes.

• Implement cryptographic encoding (Binary & HEX).

• Provide modular arithmetic and inverse calculations.

• Create interactive logic gate and matrix simulators.

System Features

Crypto Lab: Binary conversion, HEX conversion, Simple hashing.

Brute Force PIN Simulator: Demonstrates PIN cracking attempts.

Mathematical Tools: GCD, Factorial, Average, Cube Root, Percentage.

Modular Arithmetic: Modular exponentiation and inverse.

Probability Module: Basic and binomial probability.

Logic Gates Simulator: AND, OR, NOT, NAND, NOR XOR, XNOR.

Matrix Operations: Addition, Subtraction, Multiplication, Transpose.

Technologies Used

HTML – Structure

CSS – Styling and Interface Design

JavaScript – Functional Logic and Calculations

Sustainable Development Goals (SDGs)

SDG 4 – Quality Education

SDG 9 – Industry, Innovation, and Infrastructure

SDG 16 – Peace, Justice, and Strong Institutions

Conclusion

Cyber Utility is a multifunctional cybersecurity learning platform that integrates cryptographic and mathematical tools into a single interactive system. The project enhances practical learning and demonstrates innovation in cybersecurity education.